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1 What is WPI

* The Worcester Polytechnic Institute is a private research university in Worcester, Massachusetts, United States. Founded in 1865, WPI was one of the United States' first engineering and technology universities and now has 14 academic departments with over 50 bachelor's, master's, and Ph.D. degree programs.

The FIRST Robotics Competition (FRC) at WPI ([Worcester Polytechnic Institute](https://www.wpi.edu/)) refers to WPI's deep involvement in hosting and supporting the national FRC program for high school students, where teams build and compete with industrial-sized robots, fostering STEM skills, teamwork, and leadership, with WPI providing resources like their WARP preseason event and hosting competitions. It's a hands-on experience where students learn engineering, programming, and business, guided by mentors, all while aiming for scholarships and championship events.

2 Git installs

2.1 Create GitHub account

Signing up for a new personal

1. Navigate to <https://github.com/>.
2. Click **Sign up**.
3. Alternatively, click on **Continue with Google** to sign up using social login.
4. Follow the prompts to create your personal account.

During signing up, you'll be asked to verify your email address. Without a verified email address, you won't be able to complete some basic GitHub tasks, such as creating a repository.

Getting started: <https://docs.github.com/en/get-started/start-your-journey/creating-an-account-on-github>

Tutorial: [How to Create a GitHub Account](https://www.youtube.com/watch?v=RnWDKWmaQ8s) : <https://www.youtube.com/watch?v=RnWDKWmaQ8s>

2.2 Download and install

1> Git and Git Bash command line interface.

Download and install: <https://git-scm.com/>

Tutorial to install Git: <https://www.youtube.com/watch?v=tRZGeaHPoaw>

The Ultimate GitHub SSH Key Tutorial: <https://www.youtube.com/watch?v=6U53fsMiPm0&t=7s>

Generate ssh key commands: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>

Getting started: <https://docs.github.com/en/get-started/learning-to-code/getting-started-with-git>

Working with Git and GitHub: <https://www.youtube.com/watch?v=d3FkIA9RtlA>

OR

2> GitHub Desktop Client

Instead of using Git from command line, GitHub Desktop Client allows us to use Git from graphical interface. If you have GitHub Desktop installed, you can use it to clone repositories and not deal with SSH keys.

Download and install: <https://github.com/apps/desktop>

Tutorial link: <https://www.youtube.com/watch?v=8Dd7KRpKeaE>

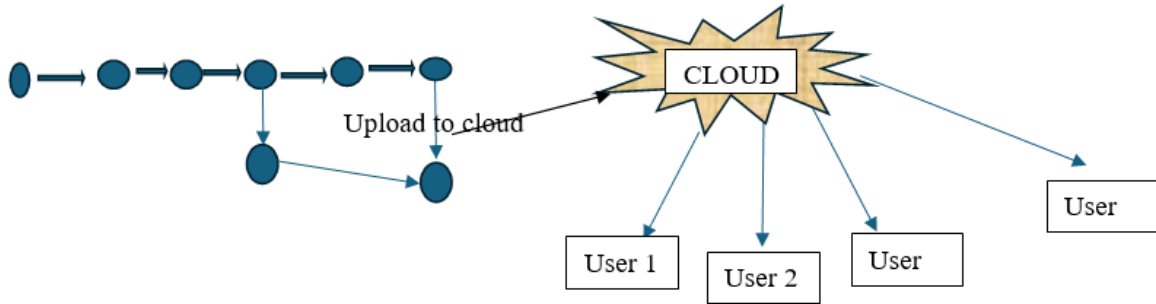
<https://www.youtube.com/watch?v=MaqVvXv6zrU>

Getting started: <https://docs.github.com/en/get-started/learning-to-code/getting-started-with-git#learning-git-basics-with-github-desktop>

Git and GitHub

<u>Git</u>	<u>Github</u>
Git is the version control software that runs locally on your machine. You don't need to register for an account. You don't need the internet to use it. You can use Git without ever touching Github	Github is a service that hosts Git repositories in the cloud and makes it easier to collaborate with other people. You do need to sign up for an account to use Github. It's an online place to share work that is done using Git.

From the previous example, all these are all on local machines. What if the local machine is lost or goes bad. All work is lost. I can upload my history to cloud and recover from the cloud. Using Github does the same job. The other advantage is that user/owner of work can share with others in the team as well.



3 Working with Git Bash commands

Getting Started: <https://typoapp.io/blog/git-bash-commands>

To get and use Git Bash inside GitHub Desktop, you need to **download Git for Windows separately** and then select it as your default shell inside the GitHub Desktop settings.

Follow this quick setup process to link them together:

Step 1: Install Git Bash on Your PC

1. Go to the official [Git for Windows Download Page](#).
2. Click **Download** to get the installer package.
3. Run the downloaded installer file (exe).
4. Click **Next** through the setup prompt windows to accept the default settings, then click **Install**.

Step 2: Set Git Bash as Default in GitHub Desktop

1. Open your **GitHub Desktop** client.
2. In the top-left menu bar, click on **File** and select **Options** (on Mac, click *GitHub Desktop > Settings*).
3. Click on the **Integrations** tab on the left sidebar.
4. Look for the **Shell** dropdown menu.
5. Select **Git Bash** from the list of available terminal choices.
6. Click **Save** to apply the configuration.

Step 3: Open Git Bash Instantly

Once configured, you can launch Git Bash directly from your active project repository using either of these quick methods:

- **The Shortcut:** Press **Ctrl + `** (Control + Backtick) on your keyboard.
- **The Menu:** Click on **Repository** in the top menu bar, then click **Open in Git Bash**.

4 VS code and Git

Tutorial: How To Use Git In VS Code Like A Pro!

<https://www.youtube.com/watch?v=twSYxYaQikI>

5 Getting Started

Introduction — FIRST Robotics Competition documentation

<https://docs.wpilib.org/en/2020/docs/getting-started/getting-started-frc-control-system/intro.html>

6 System Installation

FGR Startup notes (link): Byting Irish Programming 2023 - HackMD

<https://hackmd.io/ZvN70XOzRyG9hFiyX5rgEA#System-Installation>

7 WPI Library

More info: WPILib <https://wpilib.org/>

The WPI Robotics Library (WPILib) is the standard software library provided for teams to write code for their FRC® robots. WPILib contains a set of useful classes and subroutines for interfacing with various parts of the FRC control system (such as sensors, motor controllers, and the driver station), as well as an assortment of other utility functions.

7.1 How to install WPLib

Download and Install WPILib from

WPILib Installation Guide — FIRST Robotics Competition documentation

https://docs.wpilib.org/en/latest/docs/zero-to-robot/step-2/wpilib-setup.html?_gl=1*18rjx7*_ga*MzIzNDY2MTg2LjE3NjQ5NjAwMjA.*_ga_RE35PKQB7J*czE3NjQ5NjAwMTk0bzEkZzEkdDE3NjQ5NjAxNTgkajYwJGwwJGgw*_gcl_au*MTEwNTA2ODEyNy4xNzY0OTYwMDE5

From the above steps followed

WPLib comes with customized VSCode. It is not required to uninstall if another VSCode exist until SYNC is disabled.

The WPILib extension provides the FRC® specific functionality related to creating projects and project components, building, and downloading code to the roboRIO and more. You can access the WPILib commands one of two ways:

- By typing “WPILib” into the Command Palette
- By clicking on the WPILib icon in the top right of most windows. This will open the Command Palette with “WPILib” pre-entered

Here is link to understand the WPLib's features/WPI Extensions/Commands/Create project etc in VSCode.

Link : Visual Studio Code Basics and the WPLib Extension — FIRST Robotics Competition documentation <https://docs.wpilib.org/en/stable/docs/software/vscode-overview/vscode-basics.html#visual-studio-code-basics-and-the-wpilib-extension>

What is Installed?

The Offline Installer installs the following components:

- **Visual Studio Code** - The supported IDE for 2019 and later robot code development. The offline installer sets up a separate copy of VS Code for WPILib development, even if you already have VS Code on your machine. This is done because some of the settings that make the WPILib setup work may break existing workflows if you use VS Code for other projects.
- **C++ Compiler** - The toolchains for building C++ code for the roboRIO
- **Gradle** - The specific version of Gradle used for building/deploying C++ or Java robot code
- **Java JDK/JRE** - A specific version of the Java JDK/JRE that is used to build Java robot code and to run any of the Java based Tools (Dashboards, etc.). This exists side by side with any existing JDK installs and does not overwrite the JAVA_HOME variable
- **WPILib Tools** - SmartDashboard, Shuffleboard, RobotBuilder, OutlineViewer, PathWeaver, Glass, SysId, Data Log Tool, roboRIO Team Number Setter, AdvantageScope, Elastic, WPICal
- **WPILib Dependencies** - OpenCV, etc.
- **VS Code Extensions** - WPILib and Java/C++/Python extensions for robot code development in VS Code
- **Documentation** - Offline copies of this frc-docs documentation and Java/C++/Python APIs

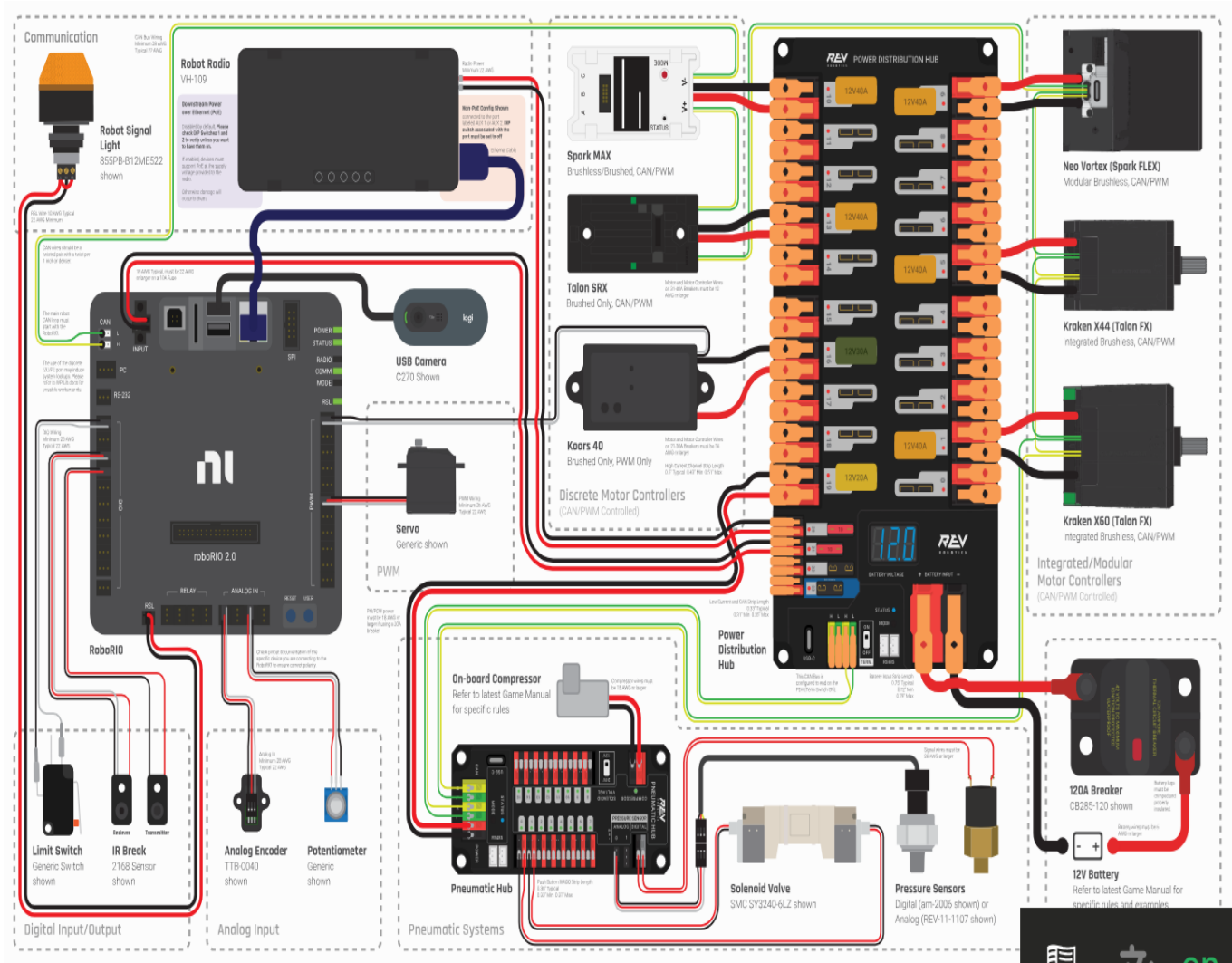
Note

It's not recommended to use the VS Code Backup and Settings Sync feature to sync settings between a regular VS Code installation and the FRC VS Code installation. This could lead to either installation being broken.

8 Hardware introduction

Hardware Component Overview — FIRST Robotics Competition documentation

<https://docs.wpilib.org/en/stable/docs/controls-overviews/control-system-hardware.html>



9 Software components for 2025

List of software components here - Software Component Overview — FIRST Robotics Competition documentation <https://docs.wpilib.org/en/stable/docs/controls-overviews/control-system-software.html>

Installing guide of software - Step 2: Installing Software — FIRST Robotics Competition documentation <https://docs.wpilib.org/en/stable/docs/zero-to-robot/step-2/index.html>

9.1 Download and install Gaming tool

Installing the FRC Game Tools — FIRST Robotics Competition documentation <https://docs.wpilib.org/en/stable/docs/zero-to-robot/step-2/frc-game-tools.html>

The FRC® Game Tools contains the following software components:

- LabVIEW Update
- FRC Driver Station
- FRC roboRIO Imaging Tool and Images

9.2 FRC Driver Station

Understand more on Driver station here

FRC Driver Station Powered by NI LabVIEW — FIRST Robotics Competition documentation <https://docs.wpilib.org/en/stable/docs/software/driverstation/driver-station.html#frc-driver-station-powered-by-ni-labview>

9.3 VSCode comes with WpiLib

9.4 LabView (optional)

- What is NI LabVIEW?
- Basics of LabVIEW Programming Environment Tutorial
- Beginners LabVIEW Tutorial 1: Getting Started with LabVIEW

9.5 Python (optional)

Install python 3.13 from microsoft store and install RobotPy package
(`$python3 -m pip install robotpy`)

Python Installation Guide — FIRST Robotics Competition documentation

10 Robo code 2025

Clone: FRC7660/robot-2025 <https://github.com/FRC7660/robot-2025>

`<path>$ git clone https://github.com/FRC7660/robot-2025.git`

11 Imaging RoboRio and setting Team number

Imaging your roboRIO 2 — FIRST Robotics Competition documentation

<https://docs.wpilib.org/en/stable/docs/zero-to-robot/step-3/roborio2-imaging.html>

12 Building and Deploying Robot Code

For more details on how to build and deploy the code on RoboRio, Check this link

Building and Deploying Robot Code — FIRST Robotics Competition documentation

<https://docs.wpilib.org/en/stable/docs/software/vscode-overview/deploying-robot-code.html>

13 Programming your Robot

Creating your Test Drivetrain Program (Java/C++/Python) — FIRST Robotics Competition documentation

<https://docs.wpilib.org/en/stable/docs/zero-to-robot/step-4/creating-test-drivetrain-program-cpp-java-python.html>

14 Running your Test Program

Running your Test Program — FIRST Robotics Competition documentation

<https://docs.wpilib.org/en/stable/docs/zero-to-robot/step-4/running-test-program.html>